a Localized climate risks to facility-level production

Climate hazards (facility level)

- **Heatwaves** — High temperatures
- **Tropical cyclones** — Strong winds and flooding
- **Wildfires** — Fire damage and smoke

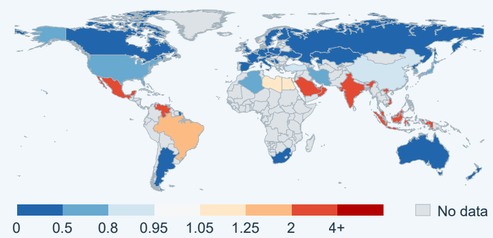
Steel production facilities

- Iron mines
- Ironmaking facilities
- Steel plants

Facility-level production losses



Country-stage production constraints



Climate-induced capacity constraints

b 2023 steel material-flow network

Global physical flows and major production facilities

Iron mines (409)

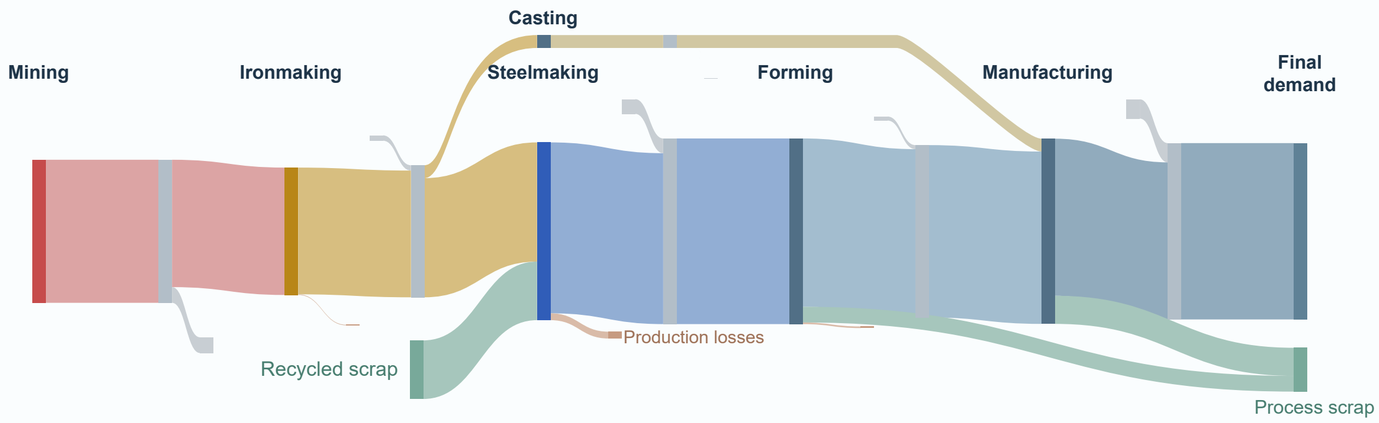
- **Hamersley Consolidated** — AUS · 138.2 Mt
- **Northern System** — BRA · 82.1 Mt

Ironmaking facilities (474)

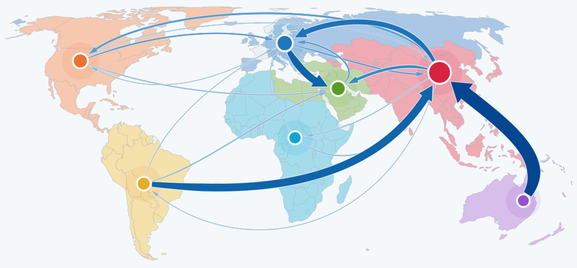
- **POSCO Gwangyang** — KOR · 19.6 Mt
- **Zhangjiagang Hongchang** — CHN · 16.9 Mt

Steel plants (871)

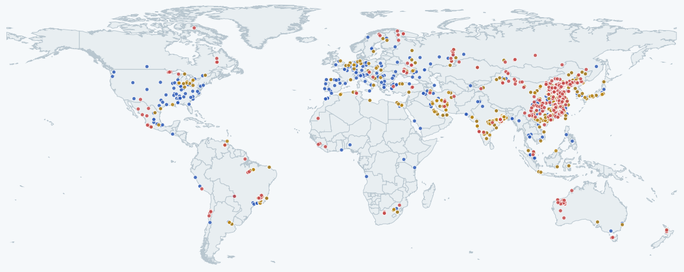
- **Angang** — CHN · 19.3 Mt
- **POSCO Gwangyang** — KOR · 18.8 Mt



Country-level trade



Locations of modelled facilities



Material flow network

c Dynamic supply-chain simulation

Agent-based model (ABM)

- **Inventory buffer** — Scenario setting
- **Trade restrictions** — Scenario setting

Production → **Inventories** → **Trade matching** → **Final supply**

State update · Δt = 7 days

Country-level final supply

Simulated final supply

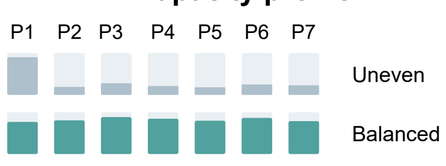
Baseline capacity and final demand

2023 demand

d National steel security and comparative analyses

Baseline capacity and final demand → **Domestic supply chain capability index (DSCCI)** — Adequacy × balance — Association — **Final-demand satisfaction**

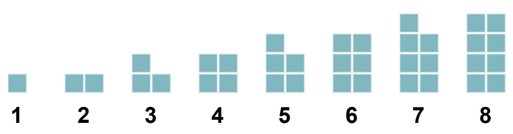
Capacity profile



Climate pathways

- Pre-industrial
- SSP1-2.6
- SSP3-7.0
- SSP5-8.5

Inventory buffers (weeks)



Trade access

